A RESEARCH PROPOSAL ENTITLED:

THE CSETI* EARTH AND NEAR EARTH EXTRATERRESTRIAL SURVEY

* Center for the Study of Extraterrestrial Intelligence

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Submitted to:

Total Amount Requested: **$ 9,429,702**

Date Submitted:

Funding Period: 3 Years from the time of receipt of funding

THE CSETI EARTH/NEAR EARTH EXTRATERRESTRIAL SURVEY

EXECUTIVE SUMMARY

Extraterrestrial civilizations which may be thousands of years ahead of us technologically would not be expected to communicate using twentieth century Earth technologies. Therefore, an alternative approach to searching the radio frequency (RF) spectrum for intelligent signals elsewhere in the universe must be considered. Central to this hypothesis is the concept that any sufficiently advanced civilizations attempting interstellar communication and travel will, perforce, develop "transluminal" (faster than light) technologies, which permit real time communication and travel beyond the speed of light (EM) barrier. The CSETI retrospective review of scientific cases, photographs, video tapes, top secret government documents and reported landing trace cases suggests that a phenomenon exists in near Earth proximity which could constitute extraterrestrial monitoring and reconnaissance of Earth. CSETI proposes a two-phase, three-year scientific research project which would conduct a global real time survey of this phenomenon and, for the first time, scientifically attempt to measure it.

INTRODUCTION AND BACKGROUND

Recent scientific research has verified the existence of a number of star systems with planetary bodies. The discovery of these extrasolar planets has increased dramatically within the past several years, so that now nearly 40 confirmed planets have been discovered. Over 500 scientific papers and reports on extrasolar planets and extraterrestrial life issues have been listed on the extrasolar planets web site just since 1998 (Extrasolar 1999). This research has also been the subject of many scientific articles in the popular press (Begley 1998; Lemonick 1996). In addition to planets, the existence of organic molecules in space which may provide the essential components necessary for life have also been reported.

According to the Drake formula (Drake 1963), it is likely that thousands of intelligent life forms have evolved within our own galaxy alone, which is only one of billions of such galaxies. The question then presents itself of whether or not intelligent life actually exists outside of Earth and, if so, how such life might manifest itself. This topic has been the subject of intense speculation over the years in both the scientific literature (Goldsmith and Owen 1980; Deardorff 1986; Tough 1999) and the popular scientific press. It is a commonly held view that somewhere, in a universe containing billions of galaxies each with billions of star systems, higher intelligent life forms are likely to have evolved.

The SETI project (Search for Extraterrestrial Intelligence) has focused on surveying space for non-Earth originating electromagnetic (EM) signals using radio telescopes and a billion-channel computerized monitoring system. The assumption used by SETI is that intelligent, artificial and repeating EM signals originating from outside of Earth or human derived near Earth technologies would be evidence of other intelligent life in the universe. While the SETI effort represents one rational approach to the question of discovering intelligent life in the universe, there are additional strategies which are herein presented which will augment the SETI project and test a number of hypotheses which are related to, but separate from, those of SETI.

In the consideration of the evolution of higher intelligent life forms, frequently omitted are questions of how such life forms might utilize laws of physics not currently known to twentieth century Earth scientists, and how such a utilization may appear to humans that encounter such technologies. While it is possible that EM signals may be detected from civilizations using those technologies, it is equally likely that such technologies of extraterrestrial origin may be utilized in a manner very foreign to humans. Moreover, it is possible, if not likely, that there are communication and transportation technologies which utilize laws of the universe, physics and applied technologies which altogether transcend current human understanding. Deardorff (1987) discusses the implications of some of these ideas.

In this regard, it is important to consider that the universe (as we currently understand it) is 15-20 billion years old. In a universe so vast and ancient it is likely that, if there are intelligent life forms outside of the Earth, they are at points of technological and social evolution different from our own. Moreover, once a civilization reaches a certain point in social and technological evolution it is likely that their further development becomes virtually quantum in nature.

Considering how rapidly society and technology have changed and evolved since the human scientific revolution, and extrapolating such exponential growth into the future 10,000 years, one can see that the technology of that future time would seem virtually like magic to humans of today. Thus, as we consider the likelihood of extraterrestrial intelligent life, it is important to avoid an overly anthropocentric view of such life, and to remember how quickly technological change may happen and how bizarre to us it may appear. Consequently, any civilization sufficiently advanced to travel through interstellar space would likely possess technologies beyond our comprehension.

This then brings us to a fundamental dilemma - advanced extraterrestrial life forms may very well utilize laws of physics and technologies which are completely beyond the known EM spectrum. Hence, if we are looking only within that spectrum, as SETI is doing, we are likely to altogether miss the existence of such civilizations.

In summary then, it may be that as intelligent civilizations evolve technologically, other spectra are discovered, mastered and utilized which altogether bypass (or transcend) the currently known EM spectrum. It is possible - if not likely - that the current human use of the EM spectrum is a technological phase of relatively short duration (like the use of fossil fuels for energy), and that within a few hundred years other spectra will be utilized for communications. Our searching for advanced extraterrestrial civilizations in the EM spectrum could be the equivalent of a civilization at the evolutionary level of cavemen looking for our civilization by trying to detect smoke rings rising from the jungle.

PROJECT RATIONALE

Extraterrestrial civilizations which may be thousands to hundreds of thousand of years ahead of us technologically will, by definition, not be using twentieth century Earth technologies. Today's scientists insist that the speed of light cannot be exceeded, but we must remember that only a few decades ago the technology that could enable an aircraft to exceed the sound barrier did not exist. Perhaps advanced extraterrestrial civilizations capable of interstellar communication and travel have evolved technologies which can exceed the speed of light or effectively work around it. While this may seem impossible to us today, the world of today would seem impossible to someone a brief 200 years ago.

Therefore, an alternative hypothesis in the search for intelligent life elsewhere in the universe must be considered. If sufficiently advanced beyond current Earth technologies, extraterrestrial life forms may be capable of interstellar travel and communications at multiples of the speed of light, or in altogether non-linear modes, which simple EM monitoring will never detect.

Central to this hypothesis is the concept that any sufficiently advanced civilizations attempting interstellar communication and travel will, perforce, develop "transluminal" (faster than light) technologies which permit real time communication and travel beyond the speed of light (or EM) barrier. As many scientists have pointed out, interstellar space is so vast that even the speed of light is too slow to accommodate communication and travel across it. A hundred light years is virtually in our galactic neighborhood, and yet at the speed of light, a 'round-trip' conversation needed for communication will take 200 years; that is, one would be deceased before the 'conversation' could be completed.

The real dilemma is the limitation of current human technologies. If we cannot project communications at multiples of the speed of light, nor receive them, then there is a low probability of detecting ET civilizations using current technologies. This dilemma is in part rooted in the arbitrary assumption that, should extraterrestrial civilizations exist, they are only 'out there'. However, it is also possible that they are also already 'here'.

The hypothesis that we are not alone and, in fact, may already be receiving visits from advanced extraterrestrial civilizations cannot be dismissed out of hand. Human civilization could be visited by advanced ET life forms and be missing the events due to anthropocentric views of technology and scientific prejudices regarding what may not be possible in terms of "transluminal" space travel and communications. Human history, and that of modern science in particular, is rife with examples of such myopia. A review of scientific, military and intelligence cases dealing with extraordinary aerial phenomena would suggest that a near Earth survey may be of more value than electromagnetically scanning the distant galaxies.

The CSETI retrospective review of scientific cases, photographs, video tapes, top secret government documents and reported landing trace cases would suggest that a phenomenon exists in near Earth proximity which could constitute extraterrestrial monitoring and reconnaissance of the Earth (see Appendix A). Moreover, the identification and interviews by CSETI of over 100 government, military, intelligence and aerospace employees, with high level security clearances and have worked in the "black world" of super secret projects, strongly suggests that there is an ongoing extraterrestrial phenomenon albeit bizarre in terms of its technological manifestations (see Appendix B).

While many such reports which exist in the general culture certainly represent misidentifications of prosaic atmospheric, man-made or celestial phenomenon - if not outright hoaxes - there is a residua of very convincing data which the objective scientist cannot ignore. Included in this body of evidence are nearly 500,000 cases in the so-called UFOCAT computerized system, over 4000 landing trace cases in which unexplained craft have landed and left physical trace evidence, over 3500 military and civilian pilot reports. Many of the cases include corroborating radar confirmation and hundreds of pages of unambiguous government documents. It is important to note that this evidence comes from all over the world, not just advanced countries. There are also dozens of scientifically analyzed photographs and videotapes which show objects which cannot be explained by conventional phenomenon or human-made craft (see Appendix C).

While the interpretation of the data may be debated, there is no question that a real, scientifically measurable aerial and near Earth phenomenon exists which may constitute extraterrestrial activity.

Recently a team of scientists lead by Dr. Peter Sturrock of Stanford University, formed a panel to review the physical evidence reports of UFO phenomenon and concluded with great publicity that there was enough evidence to warrant further scientific study (Sturrock et al. 1998; Sturrock 1999). Unfortunately, due to the sensationalizing and subsequent dismissing of the subject en toto, no global scientific, civilian, independent survey of this phenomenon has ever been conducted. **In light of the substantial body of scientific evidence which now exists, the time has come for a serious scientific global survey of this unexplained Earth/near Earth activity to begin.** Even if all of the current data only suggests a 10% likelihood of extraterrestrial activity, the implications of near Earth ET activity are so profound that a serious independent (non-governmental) scientific survey is warranted.

Due to the well-documented history of official and covert governmental machinations related to this subject, it is necessary that this global Earth/near Earth extraterrestrial survey occur independent of government, military or government aerospace contractor control or influence (see Appendix D).

PROPOSED EFFORT

CSETI (a 501-C-3 scientific research organization) proposes a two phase, three-year initial scientific research project which would conduct a global real time survey of Earth/near Earth extraterrestrial activity and, for the first time, scientifically attempt to measure it. For background information on CSETI see Appendix E. Research would occur in passive/observation and active (interactive) modes as described below.

PHASE I - GLOBAL SCIENTIFIC SYMPOSIUM

Phase I of this proposal is to convene a Global Scientific Symposium to formally collect and present in one place all of the best, credible retrospective data on the subject currently available. This part of the survey will enable the current state of knowledge of the subject to be formally organized and publicly presented at a serious scientific gathering of researchers from all over the world. It will enable the real time research project described below to proceed with maximum data on past events plus the identification of areas of ongoing anomalous aerial activity which may be extraterrestrial in nature. (For a Summary of Phase I see Appendix F).

PHASE II - REAL TIME RESEARCH

The identification of credible areas of anomalous aerial activity which are either acutely or chronically persistent would allow for the most promising setting for real time, serious scientific research and observation.

Unexplained phenomena and craft near to the Earth have been reported for over 100 years. An analysis of these phenomena results in several patterns of events that have been repeating throughout much of the twentieth century. These events are described and classified as follows:

1. Aerial objects observed transiently by multiple credible observers. These objects have been seen at close range and are variably described as discs, spheres, triangles, etc., and invariably maneuver in non-aerodynamic ways. An object seen within 500 feet is termed a close encounter of the first kind (CE-1).

2. Objects or craft that have landed or have been very near the ground hovering and which have left physical trace evidence (a landing area that is indented, burned, etc.) or electromagnetic anomalies. This is termed a close encounter of the second kind (CE-2).

3. Objects or craft at which humanoid life forms have been observed. This is termed a close encounter of the third kind (CE-3).

4. Events during which observers claim to have been onboard such objects. This is termed a close encounter of the fourth kind (CE-4).

5. Interactive events with craft or objects which involve voluntary human signaling followed by deliberate communicative response from the object. This is termed a close encounter of the fifth kind (CE-5). Although a CE-5 event may sound just too bizarre to some, Dr. Richard Haines, a former NASA scientist, has just published an analysis of several 100 of such events (Haines 1999).

Importantly, any combination of the above may occur in a given geographic region persistently over a period of time, in what has been called a 'wave' of sightings. For example, persistent waves of anomalous aerial activity have been reported during the 1990s in Belgium, Mexico, parts of the USA, Brazil, Russia, Japan and the United Kingdom (see Appendix G).

An informal network of organizations and individuals studying the subject exists which allows for the rapid identification of an ongoing wave of events to which a scientific team could be deployed. Over the past 7 years, CSETI has observed that at any given time there are areas of anomalous aerial activity being reported by credible observers, pilots, the military and others to which a properly equipped scientific team could be deployed. These areas of chronic and/or acute activity, which may persist for days to weeks to years, are prime targets for scientific research and real time observation.

<u>The following nine research components are required for an effective real time survey of potential extraterrestrial activity in near Earth proximity</u>:

1. Effective global monitoring of anomalous aerial events (AAE) followed by
2. Rigorous analysis of AAE to eliminate conventional explanations (misperception of celestial activity, experimental military aircraft, hoaxes, natural aerial phenomenon. etc.) resulting in
3. Determination of genuine anomalous aerial events which are persisting, followed by
4. Deployment of a Rapid Response Team (RRT) to the area, with fully portable scientific detection, measurement and recording equipment and personnel trained in observation and interactive protocols; followed by
5. Processing and analysis of resulting data, evidence, materials and events, followed by
6. Determination of validity of RRT evidence and events:
7. IF events are deemed likely extraterrestrial in nature, then further analysis of data, evidence and events will be done by a separate, independent and impartial Scientific Review Panel (SRP).
8. IF events and data are confirmed by the SRP, then there will be notification of appropriate scientific and governmental authorities, followed by
9. Full scientific media briefing and release of data and findings to the public.

Each of the above research components are described below.

1. Global Monitoring of AAE - Currently, there are no non-covert, professionally funded and staffed entities monitoring AAE globally. However, a network of reliable researchers, scientists and organizations who monitor the phenomenon have been identified, which could be linked to a central data center. In addition, the Internet increasingly allows for rapid reporting of such events.

   The proposed Global AAE Data Center would monitor current AAE around the world via a substantial network of volunteer researchers and organizations and would also create reporting networks with the following:

   - NORAD, NRO and NSA
   - Air Traffic Control Centers (globally)
   - Select Military points of contact
   - Civilian Airline Pilots
   - NASA
   - Local and regional police forces
   - News media (both passive scanning of the media for reports and requests to media for active reporting of events)
   - Others to be identified

2. The Global AAE Data Center will be staffed with personnel who will collect reports, analyze and prioritize validity of reports, further investigate raw data as needed and identify events which warrant close scrutiny for possible RRT deployment.

   This analysis phase of collected data and reports is key since the assessment of AAE from these reports will determine whether or not the RRT is deployed. A scientific staff with expertise in the following will be needed to make this determination:

   - Atmospheric Physics
   - Astronomy
   - Data analysis and imaging analysis (photographs, videotapes, satellite images, etc.)
   - Experimental military aircraft and covert military capabilities
   - Aerospace/Orbiting debris and reentry phenomenon
   - Other

   Reports of current AAE will be analyzed and assigned a rank-order classification, which lists the event in a fashion that allows for rapid determination of the quality of the event. Events which have multiple credible witnesses (military, pilots, police, scientists, etc) and for which other corroborating evidence exists, such as landing traces, photographs, radar confirmation or material evidence, will have the highest ranking.

3. Once an AAE is analyzed, verified and authenticated, a determination will be made as to whether the event was transient and non-repeating or part of a larger pattern of ongoing and persistent activity.

For example, an AAE of high quality may be part of a chronic background pattern of activity in the general geographic area and could indicate an intensification of the phenomenon. If so, the RRT may then be considered for deployment to the site for real time research.

Ranking of these events, then, will be twofold: by quality of the event itself on a 1-10 basis (1 indicating the weakest supporting evidence and 10 the highest) and by degree of persistence of the phenomenon in that region. Persistence ranking will be as follows: T = transient, non-repeating; AP = acute persisting phenomenon (that is sudden onset but repeating and persistent phenomenon in a given geographic area); CP = chronic, persisting phenomenon (that is an event which is part of an historical area of chronic, background activity and reports of AAE); and SAP = AAE which are persistent but intermediate between chronic and acute waves of activity.

The determination of the validity of the report, its relative rank and persistence factor will determine whether or not the RRT is deployed.

4. Deployment of the Rapid Response Team (RRT) will be decided on the above factors, plus the feasibility of getting a team rapidly into the geographic region in question. This will be decided by the Project Director in consultation with the Science Director, RRT members and members of the Scientific Review Panel when appropriate.

   Given current AAE activity, it is likely that multiple RRTs will be needed for multiple, simultaneous research projects in varying geographic areas.

   Each RRT will need the following specialists:

   - Team Coordinator/Director
   - Logistics/Travel Coordinator
   - Five Field Documentation Specialists including:
     1) Equipment and technical specialist for equipment set-up and maintenance
     2) Imaging specialist(s) for photographic, videotape, film and other imaging tasks
     3) AAE identification specialists trained in early detection and observation of AAE
     4) Extraterrestrial communication specialist proficient in high-level contact protocols.
     5) Other specialties as need arises. There will be significant cross training of the needed field specialties so rapid substitutions can occur in the field as events unfold.

Equipment used on-site for the RRT:

- Magnetometers
- Spectral radiometers
- GPS
- Night-vision equipment (third generation)
- Regular, infrared, ultraviolet and starlight-scope cameras, video cameras and film
- Portable DAT (digital audio tape) recording equipment
- Electronic sensors for laser, microwave and other energy forms
- Data acquisition and transmission equipment
- Communications equipment
- Radio-transmitters for anomalous tone projection (see below)
- TV transmitters for projection of data to AAE (see below)
- Data, landing trace and material evidence collection and storage equipment
- Portable Radar Unit
- High powered light and laser signaling equipment
- Landing configuration strobes and markers
- Other items will be added based on field experience and on recommendations of the SRP

Activities of the RRT include:

- General observation and reconnaissance of the AAE area.
- Collection of information from local sources regarding times, places of recent AAE events.
- Obtaining optimal research site which is secure and in center of AAE.
- Passive observation and scientific measurement and recording of any AAEs.
- Active, peaceful engagement of AAE once it is determined that the AAE is a presumptive extraterrestrial object under intelligent control.
- Communication with extraterrestrial object and / or occupants if possible using Contact Protocols.
- Full recording of all passive and active / interactive events and collection of scientific measurements.
- Collection and safeguarding of any objects/materials of extraterrestrial origin.

Extensive preliminary field research during global CSETI CE-5 experiments over the past 8 years has enabled CSETI to develop protocols for signaling to and engaging AAE/presumptive extraterrestrial objects. While these experiments to date have been unfunded and lacked professional scientific equipment or staff, as preliminary events they have afforded CSETI remarkable opportunities for experimenting with engagement and contact protocols (see Appendix I, CSETI CE-5 experiences 1991-1997).

Since a component of the hypothesis to be tested is that these objects are under intelligent control by advanced intelligent life forms, it is important to note that in addition to passive observation and recording, protocols will be used for active

engagement, active vectoring of AAE to the research site and interactive communication, should events evolve to that point.

Vectoring protocols include the use of projected images, projected beeping tones previously recorded during AAEs (see Appendix J), high luminosity lights and lasers, and ground landing strobes.

In the event an object approaches at close range and engages the RRT in active signaling and/or communication, the event will continue to be recorded, if possible, and the RRT will advance to protocols for higher level contact with presumptive extraterrestrial life forms and/or objects. Since an event theoretically could evolve to the point of open contact and communication with non-human life forms, it is necessary that the RRT be fully cross trained in diplomatic and communication protocols so that such an event may proceed optimally.

For this reason no weapons, or objects which could be construed as weaponry, will be allowed during any RRT activation.

For classification purposes, the following Levels of Contact with AAE/Presumptive Extraterrestrial Objects will be used:

Level I Contact:	Clear and confirmed contact, which lasts less than 60 seconds.
Level II Contact:	Clear and confirmed contact which lasts more than 60 seconds and which is clearly interactive (responding to RRT interactive protocols).
Level III Contact:	Near Landing or Landing event during which object approaches at less than 500 ft and under 100 feet AGL (above ground level) altitude.
Level IV Contact:	Landing/near landing event with signaling or substantive communication lasting more than 60 seconds.
Level V Contact:	Prolonged landing and interactive event lasting over 60 seconds.
Level VI Contact:	Prolonged landing and interactive event lasting over 60 seconds with subsequent contact with or communication with observed extraterrestrial life forms.
Level VII Contact:	Event which results in one or more members of the RRT entering presumed extraterrestrial object.
Level VIII Contact:	Event which results in RRT member(s) departing with presumed extraterrestrial object and/or life forms.

Level IX Contact: Return of RRT members from Level VIII Contact above with substantive information and evidence, OR exchange of substantive information, evidence and technologies.

Every effort will be made to conduct full scientific measurements, recording and retrieval of data at every level of contact. Manual recording (still and motion picture cameras) which is not dependent on electrical or electronic energy systems will be also used since past experience with the phenomenon by CSETI CE-5 teams indicates that anomalous EM interference may render electronic devices inoperable.

Should a Level VI Contact or higher event transpire, communication of this fact will be made as soon as practical to the United Nations Secretary General, the UN Security Council and the United States President through secure points of contact already developed by CSETI.

5. Processing and analysis of data resulting from RRT events.

   Depending on the degree of contact and type of evidence recovered, a multidisciplinary team of scientists will be utilized for analysis of the event. The data from regular, infrared, ultraviolet and starlight cameras and videotape as well as electronic data will be processed and analyzed by scientists specializing in those areas.

   Should materials be retrieved which are of presumptive extraterrestrial origin, these will be secured and then analyzed by experts in metallurgy, biology, histology, etc. as warranted. Should a landing/near landing occur, the landing site will be secured and thoroughly studied and analyzed, and soil and vegetation samples will be retrieved for later laboratory analysis.

6. At the completion of the analysis of the event and all evidence, a determination will be made regarding the nature of the event and the validity and usefulness of the evidence. This final in-house determination will be made by an Event Determination Team (EDT) made up of the project directors and members of all the RRTs. If the event is regarded as extraterrestrial or very likely extraterrestrial in nature. it will be referred to scientists independent from the project.

7. The Scientific Review Panel (SRP) will consist of impartial scientists from multiple disciplines who will review all evidence, materials, recordings, measurements, etc. and perform independent analysis of it. The SRP will be asked to make an independent determination regarding the event(s) in question. Should the SRP make a determination, which differs, from the in-house scientific team's, a third analysis will be performed by another separate scientific panel. Should the event continue to be regarded as indeterminate by this team, the event will be reported publicly in a manner which reflects the differing assessments.

8. Should the SRP concur with the CSETI EDT regarding the validity of the event and make an assessment that it was likely or definitely extraterrestrial in nature, CSETI and the SRP will jointly notify appropriate scientific and governmental authorities. Entities to be notified include:

   - The UN Secretary General and UN Security Council
   - The International Astronomical Union
   - The UN Office of Outer Space Affairs
   - The US National Academy of Sciences
   - The National Science Foundation (US)
   - The Russian Academy of Sciences
   - NASA
   - Other nations' academic institutes
   - The Executive Office of the President (US) and the Science Advisor to the President

9. A full scientific media briefing will be conducted, after an appropriate interval to allow for review by the above authorities, at which the data and findings will be released to the public. A reputable and competent media relations firm will be engaged to coordinate the media briefing and subsequent media inquiries, which would be substantial.

BUDGET AND BUDGET RATIONAL

Phase I period of performance and cost: 1 Year; $80,000
Phase II period of performance and cost: 3 Years; $9,349,702

Personnel

Office Team Members

Because of the nature of the project, some of the members of the Office team may also participate in field work now and then in order to understand that aspect of the project and fill in if personnel are unable to participate.

Director - Responsible for:
- Overall project vision and goals.
- Outreach to different national/international scientific and governmental agencies
- Raising money.
- Synthesis of results in conjunction with the Science Director and other members of the project.

Science Director
- Works closely with the Director with regards to experimental design, field collection of data, and analyses of it.
- Consults with a number of outside experts (consultants), since the range of phenomena is typically beyond the scientific scope of any individual.
- Stands in for the Director at meetings and public presentations as needed

Project Manager/Financial Officer
- Works with the directors and staff to coordinate details of the field logistics and office management issues.
- In charge of all the financial issues regarding the project including oversight on spending, setting up the accounting system, and reporting needs to the funding agency.

Secretary and bookkeeper
Shared duties between the secretary and the bookkeeper will include
- Equipment and supplies ordering.
- Travel arrangement planning for the staff and teams in conjunction with each team coordinator.
- Routine book keeping.
- Answering telephones and routine office filing, etc.

AAE Scientist and Data Archivist
- Involved in the collection and scientific analysis of AAE information obtained from all over the world.
- Works directly with team members to develop methods for collection and storage of the field data.

Field Team Members

Each Rapid Response Team (RRT) will consist of seven members: the Team Director, a Field Logistics Travel Coordinator, and 5 Field Documentation Specialists. When the team members are not in the field they will be back at the main office, analyzing and synthesizing their field data. They will also aid in the collection, analysis and interpretation of AAE data from around the world.

- The **Team Director** will have overall responsibility of the scientific direction, site selection and personnel responsibilities during the fieldwork.

- The **Field Logistics Travel Coordinator** will make all arrangements for travel and housing during the field deployment and will often precede the team to the site to make these arrangements.

- The **Field Documentation Specialists** will each be in charge of various observational equipment used to collect data and will have their own specialties.

Phase-in of the field component. - Because of the potential difficulty of finding and training an adequate number of field personnel, there will only be 2 field teams during the startup year (year 1) and the 3rd team will be added during the second year.

Benefits Costs

Benefits costs are based on a percentage of the salary costs and include social security, insurance, health benefits, and retirement contributions.

Supplies and Office Rent

Supplies - Supplies include all things that are disposable or not equipment including: monthly costs for office needs including communications (phone, FAX, Internet access, AOL or equivalent accounts for field team Internet access), copier rental and copy costs, office supplies (paper, pens, disks, tapes, etc.). Estimated cost of field supplies is $500 per field trip week, including batteries, film, recording tapes, computers, disks, etc. for seven team members.

Office Rent - Rented office space must be large enough to house all the project personnel, computers, and equipment needed to run the project. The project's central offices will be based in the Charlottesville area of VA.

Equipment

Two types of equipment are listed: office and field. The office equipment includes mainly computers (desktops for personnel staying at home, staff, AAE data archiving, etc.) and office furniture. There are some extra desktop computers to be used by some of the field team personnel when at home.

Laptop computers will be available for all the field personnel and will be usually used by them when they are in the office. Extra laptops are included, since office people will need them for traveling and extras will be needed for data collection in the field. In year 3 the cost of 7 new laptops has been added to replace or upgrade those purchased in year 1.

Field equipment is divided into two main categories (Team personnel items and Team Items). Personnel items are those assigned to each field person. One to several extra of each of the personnel items per team have been included to replace lost, stolen, or broken items while in the field. The team items are those required by the whole team to collect field data and are almost entirely commercially produced reducing the need for equipment development. The costs given on the list are estimates. Lastly, one 4WD vehicle per team is necessary, to be outfitted with electronics and used for experimental work and field work on the North American continent. The second vehicle is expected to be rented at the site for the regional work. The teams will have to rent both vehicles in foreign countries.

A 10% repair and replacement cost for equipment for the life of the project has been added to cover breakage or losses.

Travel Costs

Team travel - Travel costs are based on an average 2 to 4 week field trip per team, 7 times per year, with 2 teams deployed during year 1, and 3 teams during years 2 and 3. Each team is expected to be in the field for approximately 20 weeks per year. Airline ticket costs are estimated at a high level since the teams will often travel rapidly in response to reports of major sightings around the world and thus do not have the option of purchasing tickets well in advance. Lodging and food expenses represent an average of high costs in parts of the US, Europe and Japan and lower costs in many of the third world countries.

Administrative travel - This reflects the following costs:

- Travel by Director or the Science Director to meetings either to make arrangements or present study results at scientific meetings throughout the world.
- Travel by Director or the Science Director to periodically visit the field teams.
- Annual Scientific Advisory Panel meeting to review the results of the year's work and make recommendations for future work.

Consultants

Because of the broad scope of the scientific nature of the phenomena being observed, no single individual has all the expertise needed to understand what is being observed with the AAE phenomena. Consultants or advisors are expected to be hired, especially during the first year, to help design the electronic monitoring systems. This is especially true in this field since much the expertise and knowledge is tied up in covert projects not readily available in the normal scientific literature.

Global Scientific Symposium (Phase I)

Cost estimates for the Global Scientific Symposium include approximately 40 invited speakers at an average of $1500 each (room, board, and travel) for a 4 day meeting plus 10 CSETI staff members. This totals about 50 people or about $75,000 cost for the attendees. Estimated funding for organizational expenses (room, taping, and proceedings) is $5000. The total funding for Phase I is $80,000.

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LIST OF ACRONYMS

AAE	Anomalous Aerial Events
CE-1	Close Encounter of the First Kind
CE-2	Close Encounter of the Second Kind
CE-3	Close Encounter of the Third Kind
CE-4	Close Encounter of the Fourth Kind
CE-5	Close Encounter of the Fifth Kind
CSETI	Center for the Study of Extraterrestrial Intelligence
DAT	Digital Audio Tape
EDT	Event Determination Team
EM	Electromagnetic
RF	Radio Frequency
RRT	Rapid Response Team
SETI	Search for Extraterrestrial Intelligence
SRP	Scientific Review Panel

LIST OF APPENDICES

CSETI Proposal Budget DRAFT 11/3/1999

CSETI BUDGET SUMMARY

		Year 1	Year 2	Year 3	Project Totals
Budget reflects # of field teams		2	3	3	
Personnel (#)					
Management (4)		365,000	383,250	402,413	1,150,663
Office Support staff (2)		60,000	63,000	66,150	189,150
AAE Staff (1)		50,000	52,500	55,125	157,625
Field Teams (2 or 3)		610,000	960,750	1,008,788	2,579,538
Total Personnel Salary Costs		1,085,000	1,459,500	1,532,475	4,076,975
Benefits (Soc. Sec., health insurance, retirement benefits, etc.)	35%	379,750	510,825	536,366	1,426,941
Supplies and Rent		186,500	207,075	215,554	609,129
Office Equipment and Computers		126,000	27,000	21,000	174,000
Field Equipment (Yr. 3 inc. 10% replace)		227,640	113,820	22,764	364,224
Team Travel and Field costs		499,600	749,400	749,400	1,998,400
Administrative Travel costs		52,000	52,000	52,000	156,000
Consultants		75,000	50,000	50,000	175,000
Global Scientific Symposium		80,000			80,000
TOTAL COSTS		2,631,490	3,169,620	3,179,559	8,980,669
Contingency (5% of all expenses except equip.)	5	131,575	158,481	158,978	449,033
Yearly Costs & TOTAL PROPOSED BUDGET		**2,763,065**	**3,328,101**	**3,338,537**	$\boxed{\$\ 9{,}429{,}702}$

Breakdown of Supplies and Rent Costs

Office Rent plus utilities (elect., heat, water)			120,000	126,000	132,300
Communications (Phone, FAX, Internet access, AOL)			24,000	25,200	26,460
Copier rental + copying			7,500	7,875	8,269
Office Supplies			10,000	10,500	11,025
Field Supplies (film, batteries, tapes, etc.)					
Cost per field team week in field times weeks/team	500	20	20,000	30,000	30,000
Equipment and liability insurance for field work			5,000	7,500	7,500
TOTAL SUPPLIES (including rental costs)			186,500	207,075	215,554

Radar/laser detectors w/ recording output	500	2	2000	1000	
Radio Frequency Spectrum Analyzer	25000	1	50000	25000	
Portable radar system w/recording output	5000	1	10000	5000	
Portable DAT (digital audio tape) recorders	500	2	2000	1000	
Communications equipment(Walkie talkies)	500	5	5000	2500	
Radio transmitters for tone projection	500	2	2000	1000	
Landing strobes and markers	25	6	300	150	
TV Transmitters for data to AAE (costs uncertain)		1	0	0	
Psychotronic detection devices (costs uncertain)		1			
Team Light Imaging Items					
Video camera 1 - w/ nightscope	4000	1	8000	4000	
Video camera 2 - w/ wide angle all sky lens	4000	1	8000	4000	
Video camera 3 - w/ wide range zoom lens	4000	1	8000	4000	
Windup movie camera	500	1	1000	500	
35mm mech. cameras w/ assort. lenses (Older Nikon or equiv.)	1000	2	4000	2000	
Digital still cameras	1000	1	2000	1000	
35mm quality aim/shoot cameras	350	2	1400	700	
Stabilizing binoculars high power (Cannon15x45 IS)	1400	1	2800	1400	
Spotting scopes (1 visual, 1 video)-Televue	800	2	3200	1600	
Tripods for cameras	125	6	1500	750	
Other Team Items					
Power inverters (12v to 120v)	600	1	1200	600	
Shipping Boxes (various sizes)-(Hardigg)	300	5	3000	1500	
Battery chargers	200	2	800	400	
4WD electronics vehicles for US field work	30,000	1	60000	30000	
					Year 3 10% Replace
	<u>Total Field Equip.</u>		227,640	113,820	22,764